Rishika Randev

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Education

Duke University — Durham, NC
Master of Science in Interdisciplinary Data Science — *Aug. 2024 – May 2026*

- **Honors/Awards:** Duke AI Hackathon, 2nd place (Fitting Room: an e-commerce platform for searching clothing by style)

University of Virginia — Charlottesville, VA
Bachelor of Arts in Global Public Health, Minor in Religious Studies — *Aug. 2018 – Aug. 2021*

- **Honors/Awards:** Echols Scholar, Center for Global Health Equity Scholar, Dean's List

Experience

Reliability Engineering Data Science Intern — May 2025 – May 2026
The Home Depot — *Atlanta, GA (remote)*

- Build agentic skills and prompts as part of an AIOps framework supporting reliability engineers during kubernetes cdk8s onboarding and the set up of an internal application monitoring tool
- Conduct a POC to predict the failure of four major in-store applications within 30 minute intervals by building a new set of classification models; compare various multivariate time series classification techniques to build a final model that successfully detected nearly 90% of failures ahead of time

Software Engineer 2 – Reliability Engineering — Oct. 2022 – Aug. 2024
The Home Depot — *Austin, TX (remote)*

- Used Google Cloud Platform tools like BigQuery and Stackdriver to manage infrastructure health and analyze production incidents, combining technical problem-solving with an understanding of business impact to uphold system reliability
- Deployed over 50 infrastructural changes and feature enhancements to the account experience, collaborating with software developers and product managers to improve customer satisfaction while ensuring 99.9% application availability

Projects

Cross-Modal Embeddings for Text-to-Image Retrieval & Generation in Radiology — Dec. 2025

- Implemented contrastive loss to train a ResNet-based vision encoder and BERT-based text encoder together, projecting chest X-ray radiology images and their associated clinical observations to a shared embedding space and achieving a recall@5 of 66.4% when retrieving relevant chest X-rays based on lung pathology descriptions
- Trained a conditional GAN to generate realistic chest X-ray images, using Fréchet Inception Distance to assess the difference in distributions between real and synthetic images

Fitting Room: Personalized Fashion E-Commerce Platform — Nov. 2025

- Created a personalized fashion discovery platform that helps users find clothing matching their style, body type, and preferences, using React, Flask, & SQLite
- Built a backend Pinecone vector database of real web-scraped clothing products and used text embedding retrieval via cosine similarity to recommend relevant products to users

AskSQL Text-to-SQL Converter — Dec. 2024

- Designed an easy-to-use Flask microservice for businesses to interact with inventory data by converting natural language questions into SQL queries, load tested to ensure low latency at up to 10,000 rps
- Deployed the containerized application using AWS App Runner, configuring it to interact with RDS for querying relational data, Bedrock for accessing Claude 3.5 Haiku, and Elastic Container Registry for automated re-deployment

Selected Publications

Li, J., Barry, C., Chen, J., Randev, R., Jorgensen, E., & Bent, B. (under review). When helpfulness becomes sycophancy: Sycophancy is a boundary failure between social alignment and epistemic integrity in large language models. Preprint, arXiv https://arxiv.org/abs/2605.05403.

Technical Skills

Languages: Python, R, SQL, Java, HTML/CSS
Professional Certifications: Google Cloud Platform Associate Cloud Engineer
DS/ML & Databases: PyTorch, TensorFlow, sklearn, pandas, matplotlib, seaborn, Google BigQuery, PostgreSQL, Pinecone DB, Tableau (certification in progress)
Cloud & DevOps: AWS, Google Cloud Platform, git/GitHub, Unix/Linux, Grafana, CI/CD, Docker
Other Skills: Statistical Modeling, Microsoft Excel, User-Centered Design, Technical Report Writing